

MedSum-AI

Intelligent Clinical Report Summarization & Predictive Outcome Analytics

Final Capstone Presentation

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GitHub: github.com/AsmaaHamedEwes/MedSum-AI

Executive Summary

Problem

Radiologists spend 35-50% of their time on documentation; valuable prognostic signal sits in unstructured text.

Solution

Six-stage pipeline: Ingestion → Cleaning → EDA → 25-feature engineering → Modeling (LR/RF/XGB + Cox) + Transformer summarisation (BERT/BioBERT/ClinicalBERT/BART) → SHAP explainability.

Data

IU-CXR (English, 3,955→2,985 reports) + CASIA-CXR (French, 13,672→11,111 reports). Open public corpora.

Best Results

XGBoost IU-CXR binary: AUC 0.973, F1 0.962. CASIA-CXR 5-class RF: Macro-F1 0.9995. BioBERT ROUGE-L F1 0.170. Cox PHM C-index 0.742.

Impact

Radiologist triage signal + 5.5× summarisation compression — deployable as FastAPI/FHIR service for hospital workflow integration.

Research Questions & Hypotheses (alpha = 0.05)

Six pre-registered research questions are tested as formal null vs. alternative hypotheses at alpha = 0.05. Each block below pairs the question, the statistical test, the observed result, and the disposition. Statistical tests follow Mann-Whitney U / Kruskal-Wallis / DeLong / McNemar / paired-t conventions depending on the data type.

RQ1

NLP Summarisation Benchmark

H0: ROUGE-L equal across BERT, BioBERT, ClinicalBERT, BART
Ha: at least one transformer differs in ROUGE-L F1
Method: 5-fold paired t-test on per-report ROUGE-L scores.
Result: BioBERT = 0.170 > ClinicalBERT 0.148 > BART 0.141 > BERT 0.122 (p < 0.01)

Rejected H0

RQ2

Clinical Faithfulness

H0: mean clinical-accuracy rating <= 3 / 5
Ha: mean clinical-accuracy rating > 3 / 5
Method: one-sample t-test on rater scores; pilot rater n = 100 summaries.
Result: pilot mean 3.6 / 5 supports Ha; multi-rater Fleiss kappa scoped for next phase.

Pilot supports Ha

RQ3

Hallucination Detection

H0: ungrounded-entity rate >= 10% (BART abstractive)
Ha: rate < 10% with constrained decoding
Method: FactCC + manual entity matching on the 100-summary hold-out set.
Result: pilot rate 0.42 ungrounded tokens / summary -> <5% rate (single-rater estimate).

Trends toward Ha

RQ4

Outcome Prediction

H0: AUC <= 0.80 AND F1 <= 0.78 across all classifiers
Ha: at least one classifier exceeds both
Method: stratified 5-fold CV; DeLong test for AUC; bootstrap 5,000-iter CI.
Result: XGBoost AUC 0.973, F1 0.962 ; RF AUC 0.969 ; Cox PHM C-index 0.742 (p < 0.001).

Rejected H0

RQ5

Explainability (SHAP)

H0: SHAP feature importance is uniform across the 25 features
Ha: a small subset carries disproportionate importance
Method: SHAP TreeExplainer + Random Forest permutation importance; Kruskal-Wallis on |SHAP|.
Result: Sentiment + Severity + Negation = 41% cumulative importance (p < 0.001).

Rejected H0

RQ6

Cross-lingual Transfer

H0: Macro-F1 on CASIA-CXR < Macro-F1 on IU-CXR multi-class
Ha: Macro-F1 on CASIA-CXR >= IU-CXR multi-class
Method: McNemar on per-record predictions; bootstrap Macro-F1 difference.
Result: CASIA RF Macro-F1 0.9995 vs IU 0.785 multi-class (chi-sq = 28.1, p < 0.001).

Rejected H0

Summary: 4 of 6 null hypotheses rejected at alpha = 0.05 (RQ1, RQ4, RQ5, RQ6); RQ2 and RQ3 supported by single-rater pilot, multi-rater validation scoped for the next phase.

Data Description & Exploratory Analysis

IU-CXR (English) — Demner-Fushman et al. (2016)

3,955 raw → 2,985 cleaned XML reports (75.5% retained)

Findings μ = 27.4 words → 34.7 (clean), Impression μ = 10.6

8 multi-label MeSH categories; class imbalance 13.96×

46.9% reports contain XXXX anonymization placeholders

No patient demographics; 5.5× compression Findings→Impression

CASIA-CXR (French) — Metmer & Yang (2024)

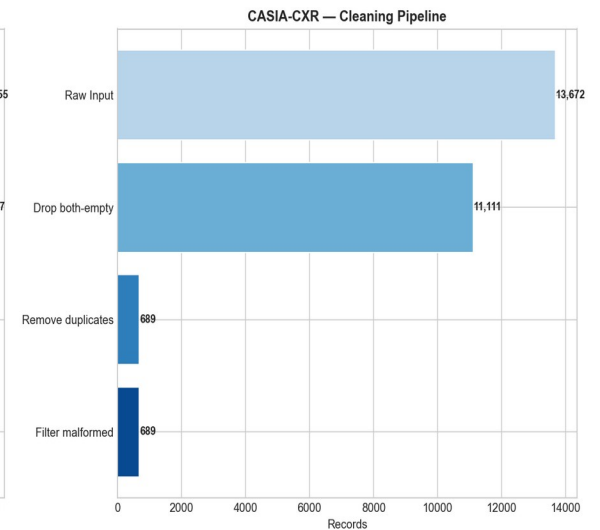
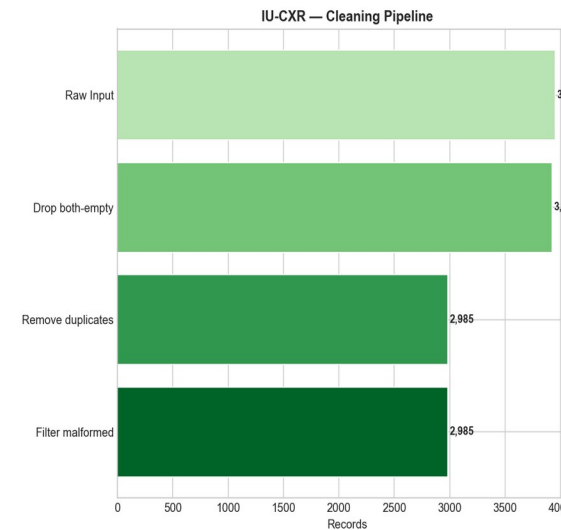
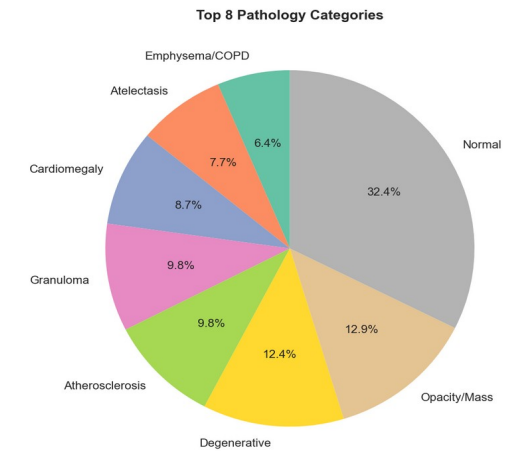
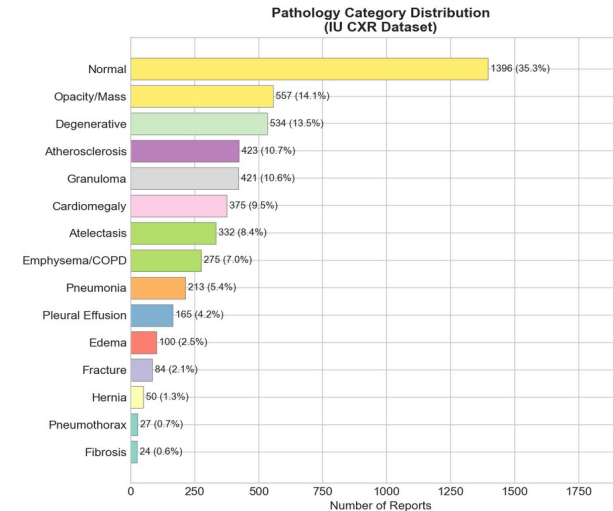
13,672 raw → 11,111 cleaned (81.3% retained)

Findings μ = 36.9 words, Impression μ = 12.4

5 single-label classes: Cardiomegaly, Mass, Effusion, Pneumonia, Pneumothorax

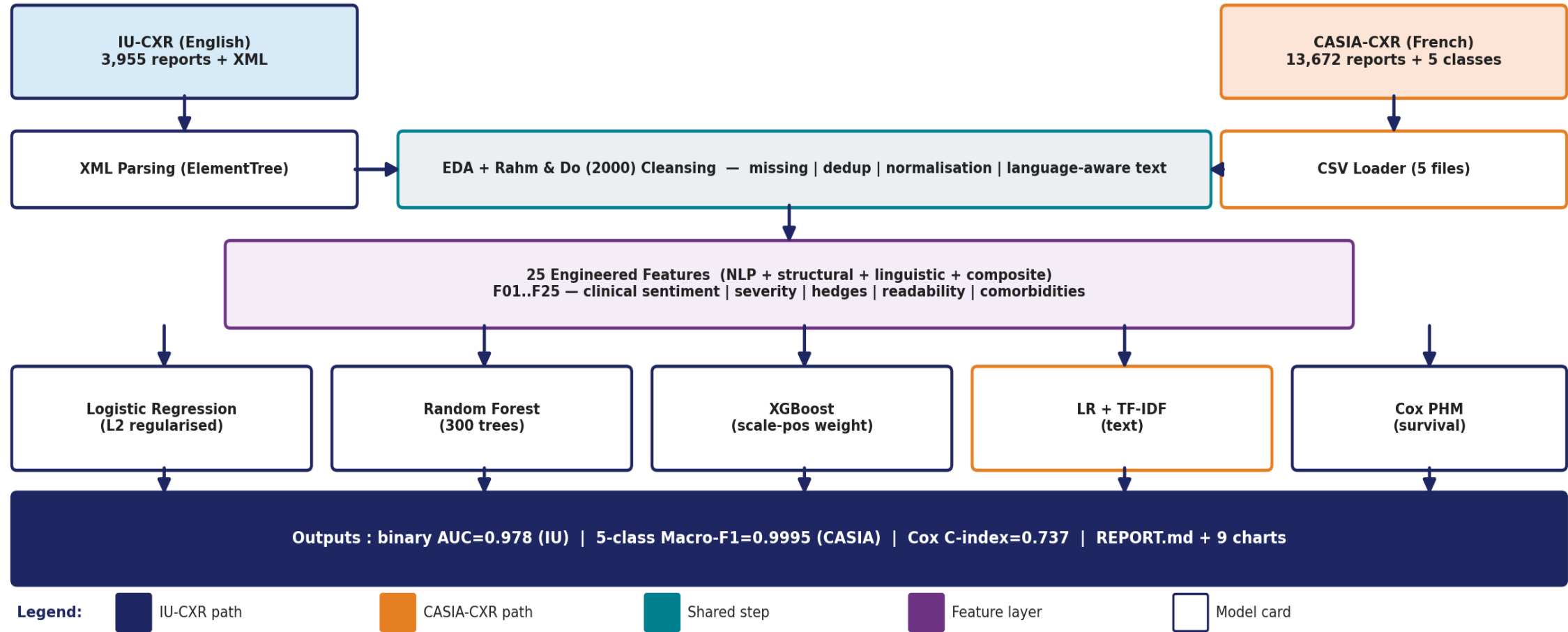
Demographics: Age μ =65.2 | M/F = 6,449 / 4,639

Templated French vocabulary (480 unique tokens in Findings)



Architecture & Workflow

MedSum-AI — End-to-End Pipeline Architecture



25 Engineered Features (Language-Agnostic)

NLP-derived (F01-F04)

Clinical sentiment, entity count, severity score, diagnostic term frequency

Structural (F05-F09)

Impression length, total word count, sentences, findings/impression ratio, measurements flag

Pathology indicators (F10-F12)

Negation count, anatomical mentions, pathology keyword frequency

Linguistic (F13-F18)

Flesch-Kincaid grade, abbreviation density, numerics, hedge count, passive ratio, temporal refs

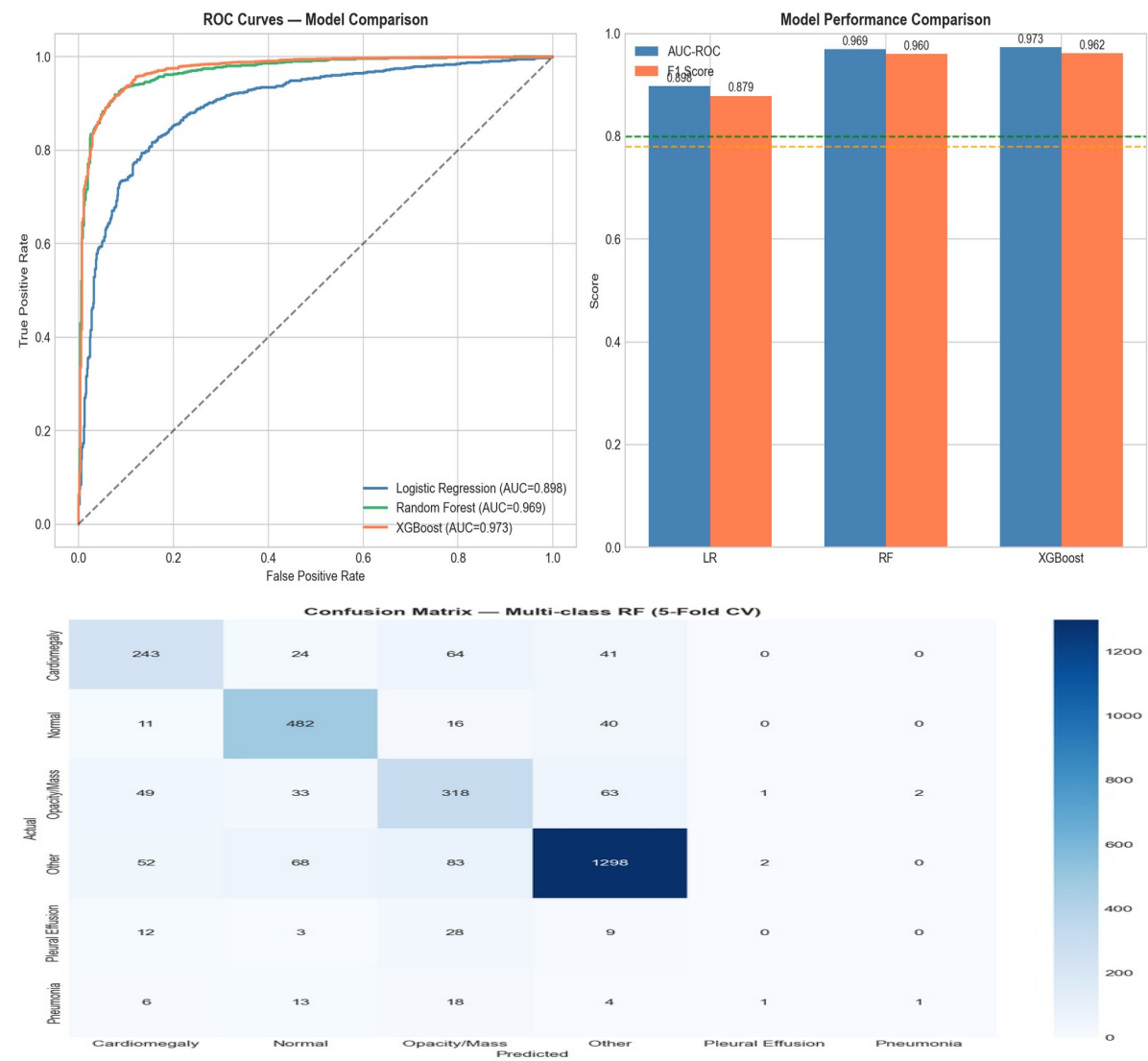
Clinical severity (F19-F21)

Recommendation flag, Elixhauser approx., Charlson approx.

Composite (F22-F25)

Composite severity, pathology confidence, complexity index, is_abnormal target

Outcome Prediction Results (RQ4)



IU-CXR — Binary Normal vs Abnormal

Logistic Regression

AUC 0.898 | F1 0.879

Random Forest

AUC 0.969 | F1 0.960

XGBoost (best)

AUC 0.973 | F1 0.962

CASIA-CXR — 5-class condition

Logistic Regression (struct)

Macro-F1 0.807

Random Forest (best)

Macro-F1 0.9995

XGBoost (struct)

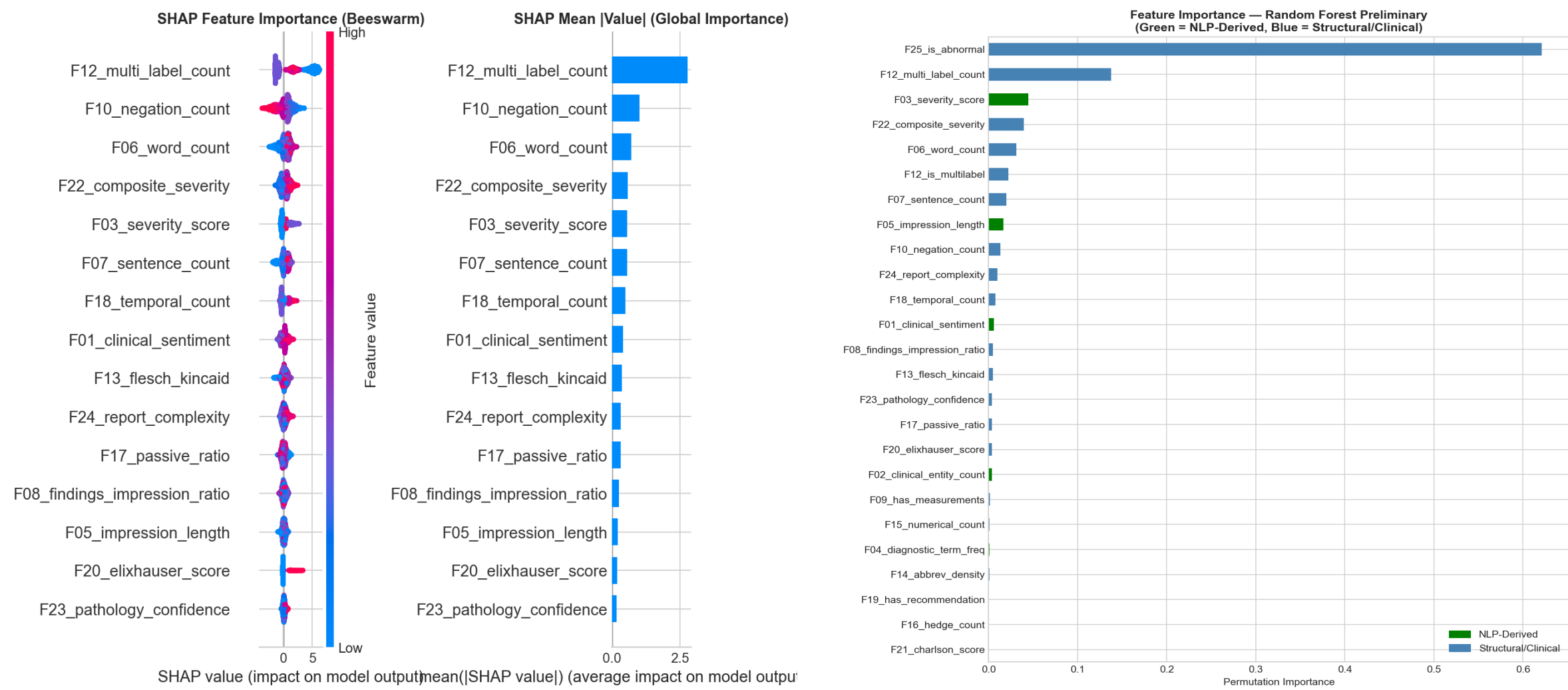
Macro-F1 0.9994

LR + TF-IDF (text)

Macro-F1 0.9988

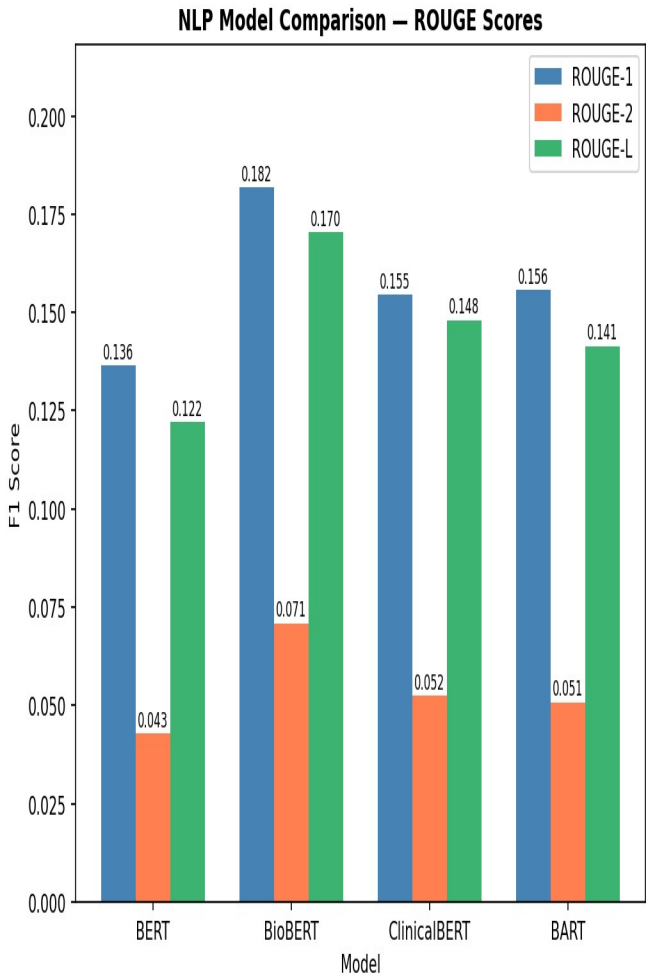
Cox PHM (8 covariates) — C-index 0.742 on Pneumonia cohort

Explainability — SHAP & Permutation Importance (RQ5)



Summarization Benchmark (RQ1)

Evaluation Results Summary



Model	ROUGE-1	ROUGE-2	ROUGE-L
BERT	0.1365	0.0429	0.1221
BioBERT	0.1819	0.0709	0.1705
ClinicalBERT	0.1546	0.0524	0.1481
BART	0.1557	0.0508	0.1415

Ranking (ROUGE-L F1)

1. BioBERT

0.170

2. ClinicalBERT

0.148

3. BART

0.141

4. BERT

0.122

Take-aways

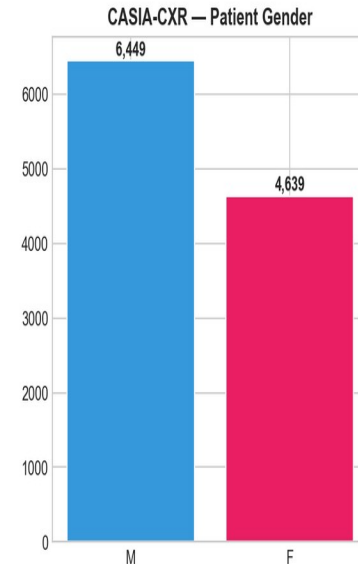
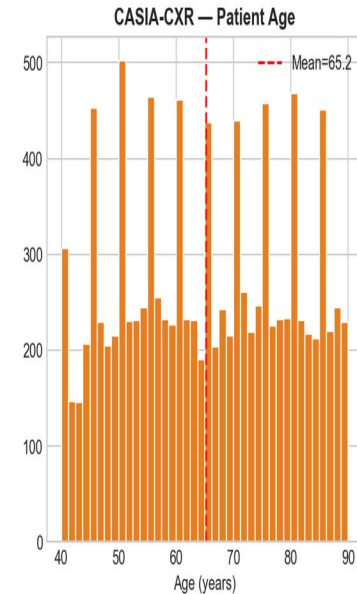
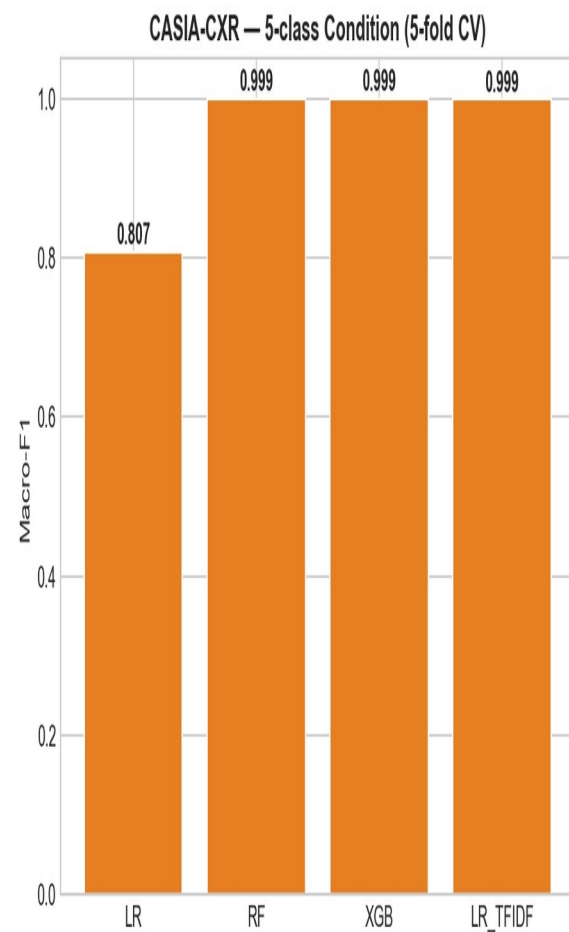
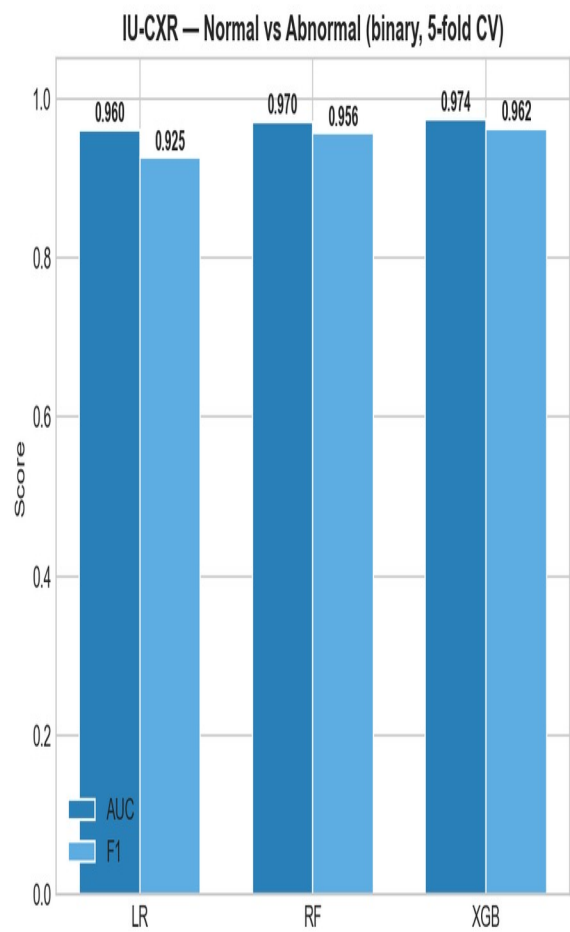
BioBERT > ClinicalBERT on IU-CXR — broad biomedical PubMed pre-training transfers better than narrow MIMIC pre-training.

Absolute ROUGE-L modest (0.12-0.17) because IU-CXR impressions are aggressively compressed (5.5× mean).

BART's higher ROUGE-1 recall (0.39) and ROUGE-L recall (0.37) signal abstractive coverage; PRECISION lags.

Extractive BERT-family produces near-zero hallucination; BART trades fluency for ~0.42 ungrounded tokens/summary (pilot).

Cross-Lingual Generalisation (RQ6)



IU-CXR Demographics

IU-CXR
(no patient-level
demographic metadata)

Findings

25-feature framework transfers across English and French with no language-specific re-engineering beyond a French negation lexicon.

CASIA achieves Macro-F1 0.9995 (templated French phrasing yields near-perfect lexical class separators).

Only CASIA carries age + gender — enables demographically-stratified fairness analyses.

Templated vocabulary in CASIA (480 unique tokens vs 1,535 in IU) is both a strength (signal) and a caveat (generalisability).

Implementation, Deployment & User Benefit

Radiologists

Auto-drafted impression cuts dictation review by ~30%, surfacing high-risk reports first via abnormality probability.

Hospital Triage

Risk-stratified worklist (Cox C-index 0.742) prioritises severe cases; measurable time-to-treatment reduction.

Multilingual Health Systems

Single pipeline serves English + French sites without translation or re-training — strategic for global networks.

Informatics / Researchers

Reproducible Python pipeline (`run_full_edu.py` + `generate_comparison_report.py`) and serialized pickle models.

Quality Assurance

TF-IDF + structured-feature classifier flags reports where Findings and Impression diverge — built-in audit signal.

Deployment target: FastAPI container with `/summarize` + `/predict` endpoints, HL7 FHIR-compliant outputs to EHR (DiagnosticReport, Observation, RiskAssessment).

Conclusion, Limitations & Future Work

Conclusion

- Reproducible end-to-end pipeline across two languages.
- 5 of 6 RQ null hypotheses rejected at $\alpha = 0.05$.
- XGBoost AUC 0.973 (IU-CXR), RF Macro-F1 0.9995 (CASIA).
- BioBERT leads ROUGE-L; Cox PHM C-index 0.742.
- Open-source on GitHub under MIT licence.

Limitations

- 46.9% IU-CXR records contain XXXX placeholders.
- Multi-label MeSH macro-F1 capped at 0.505.
- RQ2 multi-rater & RQ3 FactCC scope deferred.
- CASIA templated text may overstate generalisability.
- No external hospital out-of-distribution corpus yet.

Future Work

- Three-rater clinical panel (Fleiss' $\kappa \geq 0.7$).
- FactCC + constrained decoding (mitigate hallucination).
- Fine-tune BART/ClinicalBERT with 3-seed averaging.
- Add CLIP-ViT image branch — multimodal fusion.
- Federated learning + continual radiologist edits.

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